

SURAJ KUMAR

Phone: +91-9793330986

Email: surajk201611@gmail.com

LinkedIn: <https://www.linkedin.com/in/ksuraj1110/>

EDUCATION

Indian Institute of Technology Bombay

[June'22-May'24]

Master of Technology in Aerospace Engineering

Grade: 8.96

Indian Institute of Engineering Science and Technology Shibpur

[July'18-May'22]

Bachelor of Technology in Aerospace Engineering and Applied Mechanics

Grade: 8.04

PROFESSIONAL EXPERIENCE

MathLogic | Data Scientist

[Mar'26-present]

AI-Powered Purchase Requisition Reviewer — GenAI procurement review platform for a US-based client

- Contributed to an AI-powered purchase requisition review platform by developing scalable, reusable frontend components using **Next.js** and integrating frontend workflows with backend APIs.
- Built **GenAI document processing** with **Google Gemini** to analyze requisition descriptions, comments, and attachments, extracting procurement evidence, summarizing documents, and assisting category classification.
- Implemented a hybrid **LLM + business rules** pipeline for policy-compliant classification, validation, and exception detection, comparing AI predictions against **ERP** categories and routing only mismatched cases for human review.

Credit Risk Modeling — risk classification pipeline for portfolio decisioning

- Developed a credit risk prediction pipeline with **224+ engineered features** using **PySpark**, **Pandas**, and **Scikit-learn**, applying preprocessing, feature transformation, **IV/WOE** analysis, and correlation-based selection.
- Trained and evaluated **XGBoost**, Random Forest, and Decision Tree models using **Train/Test/OOT** validation, assessing performance with **ROC-AUC**, **KS statistic**, and **Log Loss**.

IMT – GA4 Reporting Data Warehouse & Analytics Platform — cloud analytics platform (US client)

- Developed an end-to-end **GA4** analytics pipeline using **BigQuery**, **SQL**, **Dataform**, and **Tableau**, transforming raw event data into scalable reporting marts via incremental **ETL** and automated data validation.

Feature & Model Governance (FMG) Dashboard API — ML monitoring backend for a US fintech client

- Built a **FastAPI** backend for monitoring ML models across pipelines, enabling **drift analysis**, **schema validation**, **SHAP insights**, and impact estimation with **SQLite/BigQuery** integration, pagination, filtering, dashboard summaries, and on-premises deployment support.

ITS Planners & Engineers | Associate Software Engineer

[July'24-Mar'26]

Fleet Management System — Fleet Management System to track a fleet of vehicles and manage operations

- Built scalable **React frontend modules** using **TypeScript (TSX)** with token-based authentication, role-based access control, protected routing, secure API integration, and auto-hiding of unauthorized features.
- Implemented fine-grained authorization across frontend and backend using **JWT scopes**, backend validation, and token-aware navigation, enabling role-driven functionality and user-scoped data access (e.g., depot-wise asset control).

Neptune — City-scale Transit Management Platform for planning, operations, and analytics

- Improved reporting performance using **pagination**, **indexing**, and **partitioning**, achieving $\sim 2\times$ faster APIs, $5\times$ faster CSV exports, and $\sim 2.7\times$ faster PDF generation across **1L+ GPS records**.

ReMO (ITMS) — Integrated Traffic Command & Control Platform for urban mobility

- Designed GPS Tracking and Traffic Intelligence modules using **PHP with raw MySQL**, **jQuery**, and **Leaflet**, delivering live tracking, multi-vehicle replay, POI visualization, map overlays, status filters, and PDF/CSV reporting.

SKILLS

Programming : Python, SQL, C++, JavaScript, TypeScript, PHP, HTML, CSS

GenAI & ML : LLMs, Agentic AI, LangChain, LangGraph, RAG, Fine-tuning, XGBoost, scikit-learn

Frameworks : FastAPI, React, Next.js, Celery, pandas, PySpark, SQLAlchemy, TensorFlow

Data & Cloud : BigQuery, Dataform, PostgreSQL, MySQL, SQLite, Tableau, ETL & Data Modeling

Tools : Docker, Git, Linux, Postman, Grafana, Poetry, phpMyAdmin, LaTeX

Soft Skills : Communication, Team Management, Leadership, Collaboration, Problem Solving

EXTRACURRICULAR ACTIVITIES

- Secured **2nd rank nationally** in NACDEC VI VTOL Air Ambulance, contributing to design and CFD analysis.
- Achieved **1st rank** in IIT Bombay RC Plane Competition 2022 by designing and fabricating a remote-control plane.
- Completed **Amazon ML Summer School 2023**, gaining hands-on experience in supervised & unsupervised learning.
- Served as **Cultural Secretary**, H13 Council, IIT Bombay (23–24), and **Student Companion**, ISCP, mentoring 5 first-year PG students.